DIPLOMA PROGRAMME PRACTICE · HL · SET B

Biology / Paper 1A + 1B

Worked mark scheme · 75 marks · 120 minutes across 1A and 1B

Award one mark per marking point. Credit equivalent reasoning and suitable units. Multiple-choice decisions carry one mark each.

PAPER 1A · QUESTION 1 [1]

Answer B Base pairing gives 3′-TCAGGT-5′. Reading this strand from its 5′ end gives TGGACT.

PAPER 1A · QUESTION 2 [1]

Answer A The 150 base pairs comprise 60 A–T and 90 G–C. The total is 2×60+3×90=390.

PAPER 1A · QUESTION 3 [1]

Answer A Catalysis and templating address two requirements of an early replicating system. A modern experiment establishes plausibility.

PAPER 1A · QUESTION 4 [1]

Answer A Budding incorporates host membrane lipids around the viral particle. Viral genes encode the characteristic viral structural proteins.

PAPER 1A · QUESTION 5 [1]

Answer D The first name identifies the genus. A binomial alone does not settle detailed relationships or reproductive barriers.

PAPER 1A · QUESTION 6 [1]

Answer C Distances are 0.015 and 0.060 differences per site. The rate assumption allows a relative inference, not an exact mutation history.

PAPER 1A · QUESTION 7 [1]

Answer B Temporal isolation reduces opportunities for cross-pollination even without a geographical barrier.

PAPER 1A · QUESTION 8 [1]

Answer A Equal richness can conceal loss of native species and changes in composition or evenness.

PAPER 1A · QUESTION 9 [1]

Answer C Joining 96 separate residues in one linear chain requires 95 links, each formed in a condensation reaction.

PAPER 1A · QUESTION 10 [1]

Answer B Reduction alters covalent cross-links between cysteine residues. It need not change either peptide backbone.

PAPER 1A · QUESTION 11 [1]

Answer D The favourable movement of Na^+ supplies energy for coupled uphill glucose transport.

PAPER 1A · QUESTION 12 [1]

Answer D Compartmentalization permits an acidic lumen and contains hydrolytic enzymes away from most cytosolic substrates.

PAPER 1A · QUESTION 13 [1]

Answer C Photosystems and photosynthetic electron carriers are located in thylakoid membranes.

PAPER 1A · QUESTION 14 [1]

Answer C Expanding the chamber lowers pressure inside the balloons. Air moves down the resulting pressure gradient.

PAPER 1A · QUESTION 15 [1]

Answer C Higher fetal affinity facilitates loading as dissolved oxygen diffuses across the placenta along a partial-pressure gradient.

PAPER 1A · QUESTION 16 [1]

Answer A Loading lowers water potential, water enters, and a pressure difference drives bulk flow toward unloading sites.

PAPER 1A · QUESTION 17 [1]

Answer D Ca^{2+} binds troponin and the resulting movement of tropomyosin exposes actin sites for myosin.

PAPER 1A · QUESTION 18 [1]

Answer C With similar shapes, doubling length halves area per volume, reducing the area available for heat loss per unit volume.

PAPER 1A · QUESTION 19 [1]

Answer A Loss that persists after substrate renewal and cooling supports denaturation, whereas substrate exhaustion alone would be reversed.

PAPER 1A · QUESTION 20 [1]

Answer B An allosteric change can reduce the maximum attainable rate even when substrate is plentiful. These measurements do not prove irreversible binding.

PAPER 1A · QUESTION 21 [1]

Answer B Both pathways can produce CO_2 in yeast. Oxygen uptake or ethanol production would help distinguish their contributions.

PAPER 1A · QUESTION 22 [1]

Answer A Without oxygen to accept electrons, carriers remain reduced and reoxidation of NADH slows.

PAPER 1A · QUESTION 23 [1]

Answer D Rubisco adds CO_2 to ribulose bisphosphate. The unstable six-carbon intermediate yields two three-carbon molecules.

PAPER 1A · QUESTION 24 [1]

Answer A Binding of the complex at a regulatory sequence followed by altered transcription fits intracellular receptor action.

PAPER 1A · QUESTION 25 [1]

Answer D Vesicles release transmitter on one side and receptors detect it on the other. Diffusion itself is not directional.

PAPER 1A · QUESTION 26 [1]

Answer C Na^+ channels must recover from inactivation during repolarization before a new upstroke can occur there.

PAPER 1A · QUESTION 27 [1]

Answer B The plain-block control supports transfer of an active substance from the tip. Chemical identification needs additional evidence.

PAPER 1A · QUESTION 28 [1]

Answer D Sampled area is 2.5 m^2. Density is 14 m^{-2} and the extrapolation is 14×20.

PAPER 1A · QUESTION 29 [1]

Answer B A parent strand provides the base sequence against which incoming nucleotides pair.

PAPER 1A · QUESTION 30 [1]

Answer C On the lagging template, synthesis proceeds away from the moving fork in fragments later joined by ligase.

PAPER 1A · QUESTION 31 [1]

Answer C Different exon combinations can be retained in mature transcripts, potentially yielding distinct polypeptides.

PAPER 1A · QUESTION 32 [1]

Answer A Detected cleavage at a second locus is evidence of off-target activity for that guide.

PAPER 1A · QUESTION 33 [1]

Answer A Homologues separate into different products in meiosis I. Sister chromatids separate in meiosis II.

PAPER 1A · QUESTION 34 [1]

Answer B Measuring newly synthesized transcripts tests transcription directly and distinguishes it from changes in translation or RNA stability.

PAPER 1A · QUESTION 35 [1]

Answer D Equilibrium requires equal water potentials. The positive pressure potential offsets the negative solute potential.

PAPER 1A · QUESTION 36 [1]

Answer D The LH surge triggers ovulation. The experiment distinguishes this event from preceding follicular development.

PAPER 1A · QUESTION 37 [1]

Answer D The recombinant classes total 0.20. Offspring proportions estimate recombinant gametes rather than the fraction of cells with a crossover.

PAPER 1A · QUESTION 38 [1]

Answer B Insulin promotes glucose uptake and storage in responsive tissues. Inadequate secretion slows correction of post-meal hyperglycaemia.

PAPER 1A · QUESTION 39 [1]

Answer B Reduced ADH reduces collecting-duct permeability. More water remains in the filtrate and urine becomes more dilute.

PAPER 1A · QUESTION 40 [1]

Answer C q=0.60 and p=0.40. Heterozygotes are expected at 2pq=0.48, corresponding to 240 individuals.

PAPER 1B · QUESTION 1 [8]

Testing a proposed clade

(a) Identify two shared derived characters supporting an R–S clade. [2]

1 Both R and S have G instead of ancestral A at site 1.

1 Both have T instead of ancestral C at site 2.

(b) Calculate the observed R–S and R–T distances as numbers of differing sites. State why these figures are not estimates of genome-wide percentage divergence. [3]

1 R and S differ at sites 3 and 5, so the distance is two.

1 R and T differ at sites 1, 2, 4, 5 and 6, so the distance is five.

1 The table deliberately includes only six variable positions and omits invariant sites.

(c) A second gene supports an S–T group. Give two biological or sampling reasons for the conflict and an appropriate next step. [3]

1 Chance sampling of a short sequence or repeated mutation can mislead.

1 Different gene histories, for example introgression or incomplete lineage sorting, may differ from the species history; accept another distinct valid explanation.

1 Compare multiple independent loci with broader sampling before revising the species tree.

PAPER 1B · QUESTION 2 [9]

Filament overlap during contraction

(a) Calculate percentage shortening of the sarcomere and the percentage change in its combined I-region width. [3]

1 Sarcomere shortening=(2.50−2.05)/2.50×100=18%.

1 The I regions decrease by 0.45/0.90×100=50%.

1 The smaller I regions account for the measured 0.45 µm reduction in total length.

(b) Explain how the data support sliding filaments. [3]

1 A-band length is unchanged, consistent with constant thick-filament length.

1 Thin filaments slide past thick filaments, increasing overlap.

1 Z lines are pulled closer and the regions containing only thin filaments shrink.

(c) A student observes a contracted fibre after ATP supply stops and concludes ATP is unnecessary. Evaluate that inference. [3]

1 Myosin heads can remain attached to actin when ATP is unavailable.

1 ATP binding is needed for detachment and hydrolysis helps reset the head for repeated cycling.

1 A static contracted state does not demonstrate continued ATP-independent cycling.

PAPER 1B · QUESTION 3 [9]

Interpreting light and carbon dioxide responses

(a) At light intensity 600 calculate the percentage increase on raising CO_2 and compare it with the change at intensity 100. [3]

1 At 600 the increase is 17−11=6 units.

1 Percentage increase=6/11×100≈54.5%.

1 At 100 both means are 3, so there is no observed increase.

(b) Interpret the different responses using limiting factors. [3]

1 At low intensity light delivery can limit ATP and reduced NADP production.

1 At 400 ppm, uptake plateaus between 600 and 900 even as light increases.

1 Additional CO_2 increases uptake at high light, supporting a CO_2 limitation under those conditions.

(c) Explain why an identical pair of means at intensity 100 does not prove a precisely zero effect, and identify two controls or data needed for a stronger comparison. [3]

1 Means alone omit sampling variation and measurement uncertainty.

1 Report spread or confidence intervals for independent plants (or test the difference appropriately).

1 Control leaf temperature or another relevant factor such as humidity, leaf age or water availability.

PAPER 1B · QUESTION 4 [9]

Separating transcription and RNA stability

(a) Compare the two RNA measurements before transcription is stopped. Why is transcription alone an incomplete explanation if degradation conditions were unchanged? [3]

1 New RNA doubles while steady-state mature RNA increases fourfold.

1 At a constant degradation rate a twofold production increase predicts about a twofold steady-state amount.

1 The larger change suggests altered stability or processing in addition to altered synthesis.

(b) Assume exponential decay. Estimate the half-life in each treatment using the observation that $0.71^2 \approx 0.50$. [3]

1 Untreated mRNA has a half-life of 2 h.

1 After two consecutive 2 h intervals with hormone, about $0.71^2 \approx 0.50$ remains.

1 Estimated hormone-treated half-life is approximately 4 h.

(c) Evaluate a claim that the hormone directly binds the promoter of gene X. Give one experiment that distinguishes direct promoter regulation from an indirect response. [3]

1 The data support altered RNA production and decay but do not identify a binding molecule or site.

1 Measure occupancy of the promoter by the hormone-activated receptor, comparing hormone and control conditions.

1 Mutate the proposed response sequence in a matched reporter and test whether hormone-dependent activation is lost; accept another relevant causal test.